

Chapter 16. Negative-Pressure Ventilation

Antonio Corrado; Massimo Gorini

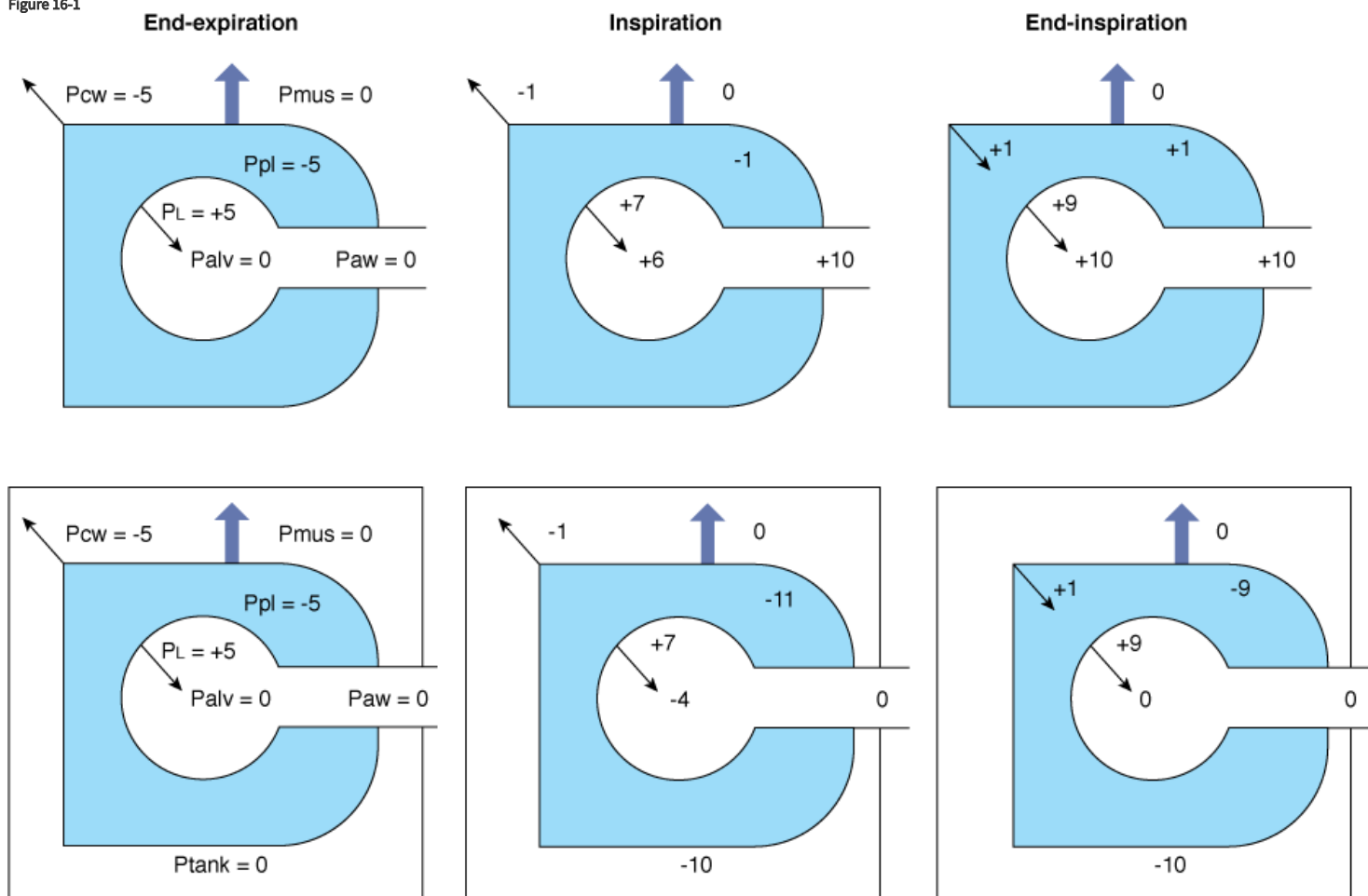
Negative-Pressure Ventilation: Introduction

Conventional mechanical ventilation via endotracheal intubation or tracheostomy in the treatment of acute respiratory failure (ARF) is a lifesaving procedure. Yet it exposes patients to severe complications, including upper airway trauma and nosocomial pneumonia, and may prolong the length of stay in the intensive care unit (ICU) and hospital because additional time may be necessary for weaning.¹⁻⁴

Basic Principles of Negative-Pressure Ventilation

Negative-pressure ventilation (NPV) works by exposing the surface of the thorax to subatmospheric pressure during inspiration. This pressure causes thoracic expansion and a decrease in pleural and alveolar pressures, creating a pressure gradient for air to move from the airway opening into the alveoli. When the pressure surrounding the thorax increases and becomes atmospheric or greater, expiration occurs passively owing to the elastic recoil of the respiratory system. The inspiratory changes with NPV, in pleural and alveolar pressures, replicate those during spontaneous breathing. On the contrary, positive-pressure ventilation (PPV) causes an increase in intrathoracic pressures during inspiration (Fig. 16-1).

Figure 16-1



Airway and intrathoracic pressures during positive-pressure ventilation (*upper panel*) and during negative-pressure ventilation (*bottom panel*). *Palv*, Alveolar pressure; *PCW*, elastic recoil pressure of the chest wall; *PL*, elastic recoil pressure of the lung; *Paw*, airway pressure; *Pmus*, muscle pressure; *Ppl*, pleural pressure; *Ptank*, tank ventilator pressure.

All NPVs have two major components: an airtight, rigid chamber that encloses the rib cage and abdomen and a pump that generates pressure changes in the chamber.^{5,6}

Negative-Pressure Ventilators

Tank Ventilator

Tank ventilators enclose the body up to the neck. The advantage is that chest wall expansion is not limited by contact with the sides of the chamber, and only one airtight seal is required around the neck. Most modern tank ventilators are constructed of aluminium and plastic and are lighter than previously. The patient's body rests on a thin mattress, and the head protrudes through a porthole at one end of the ventilator. A head and neck rest is provided in most designs to ensure comfort and to prevent upper airway collapse. Most tank ventilators have windows allowing patient observation and portholes for catheters and monitor leads and where procedures can be performed (Fig. 16-2).⁷

Figure 16-2



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com
Copyright © The McGraw-Hill Companies, Inc. All rights reserved.
Microprocessor-based iron lung (Coppa CA 1001, Coppa, Biella, Italy).

Jacket Ventilator (Pulmo-Wrap, Poncho-Wrap)

This ventilator is a windproof, water-permeable nylon parka suspended over a rigid grid that includes the rib cage and abdomen. It allows the application of negative pressure over the anterior portion of the chest wall.⁸ Airtight seals around the neck, arms, and hips are required

to prevent air leakage. The jackets do not restrain expansion of the rib cage and abdomen, but are awkward for many patients to put on and often cold to wear because of air leaks. They are preferable to tank ventilators for home use but less efficient for treating patients with ARF. The tidal volume they develop at any given level of negative pressure is less than that of a tank ventilator, and the peak pressure that patients can tolerate also usually is slightly less.⁶

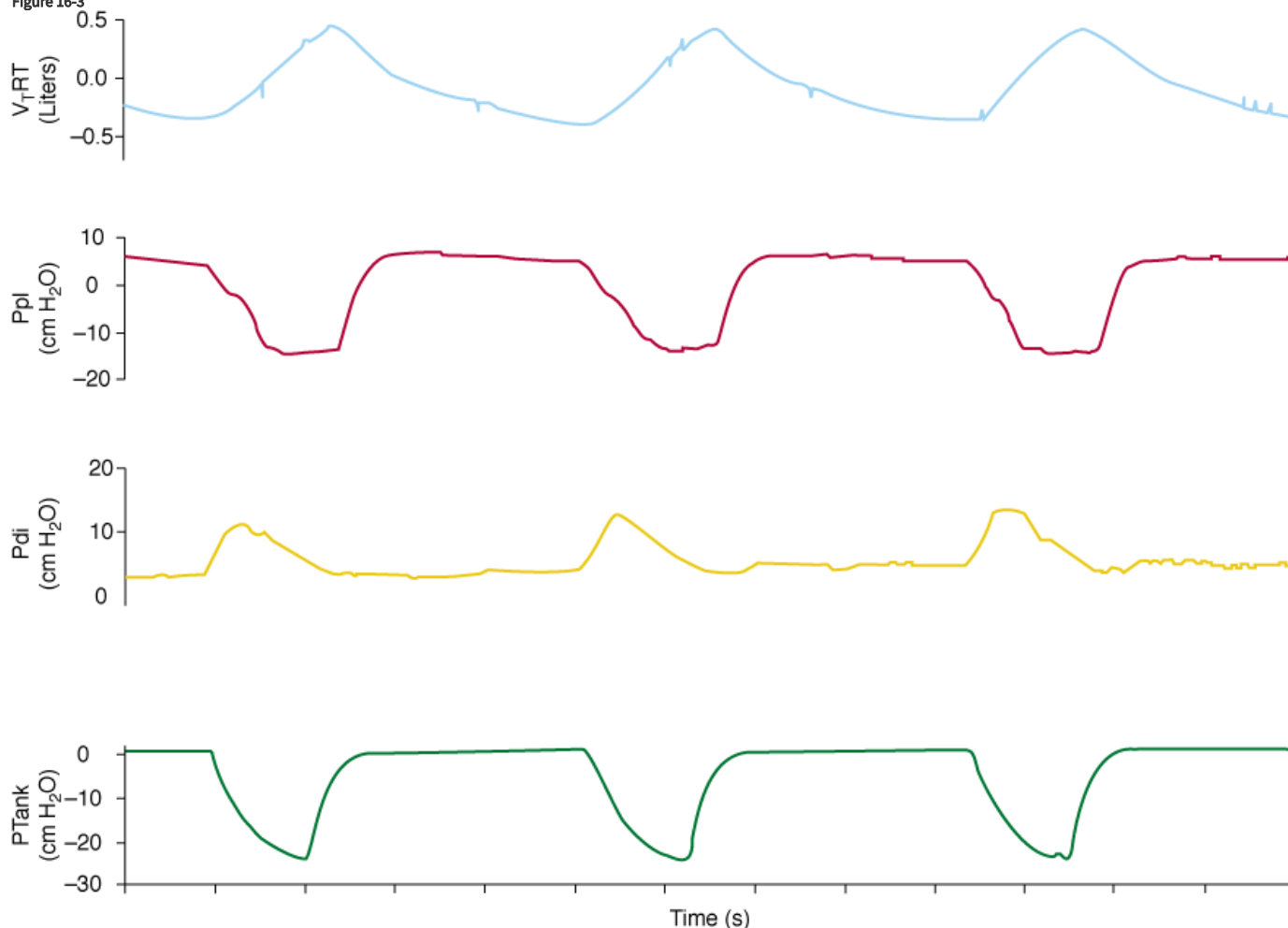
Cuirass

This consists of a rigid shell fitting firmly over the anterior portion of the chest. It applies negative pressure over a smaller surface area than either the iron lung or jacket and is the least efficient NPV.⁸ Its efficiency improves if the anterior abdominal wall is enclosed in the device and movement of the lateral aspect of the upper rib cage is not restrained. Proper fitting can be difficult, and tailor-made fiberglass shells often are necessary, particularly in patients with kyphoscoliosis.

Negative-Pressure Ventilator Pumps

Negative pressure is generated by bellows or rotary pumps that are separate or incorporated into the structure of the ventilator. Modern rotary pumps perform pressure-preset ventilation.⁹ After a breath is initiated, these pumps apply and maintain a targeted amount of subatmospheric pressure around the chest wall to meet a specified time-cycling criterion (Fig. 16-3). This mode is equivalent to pressure-controlled ventilation available with the latest PPVs. As with positive-pressure preset ventilation, tidal volume during NPV is a complex function of applied pressure and its rate of approach to target pressure, available inspiratory time, and the impedance to breathing (compliance of the respiratory system, airway resistance, and dynamic hyperinflation with intrinsic positive end-expiratory pressure [PEEP]).

Figure 16-3



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com
Copyright © The McGraw-Hill Companies, Inc. All rights reserved.

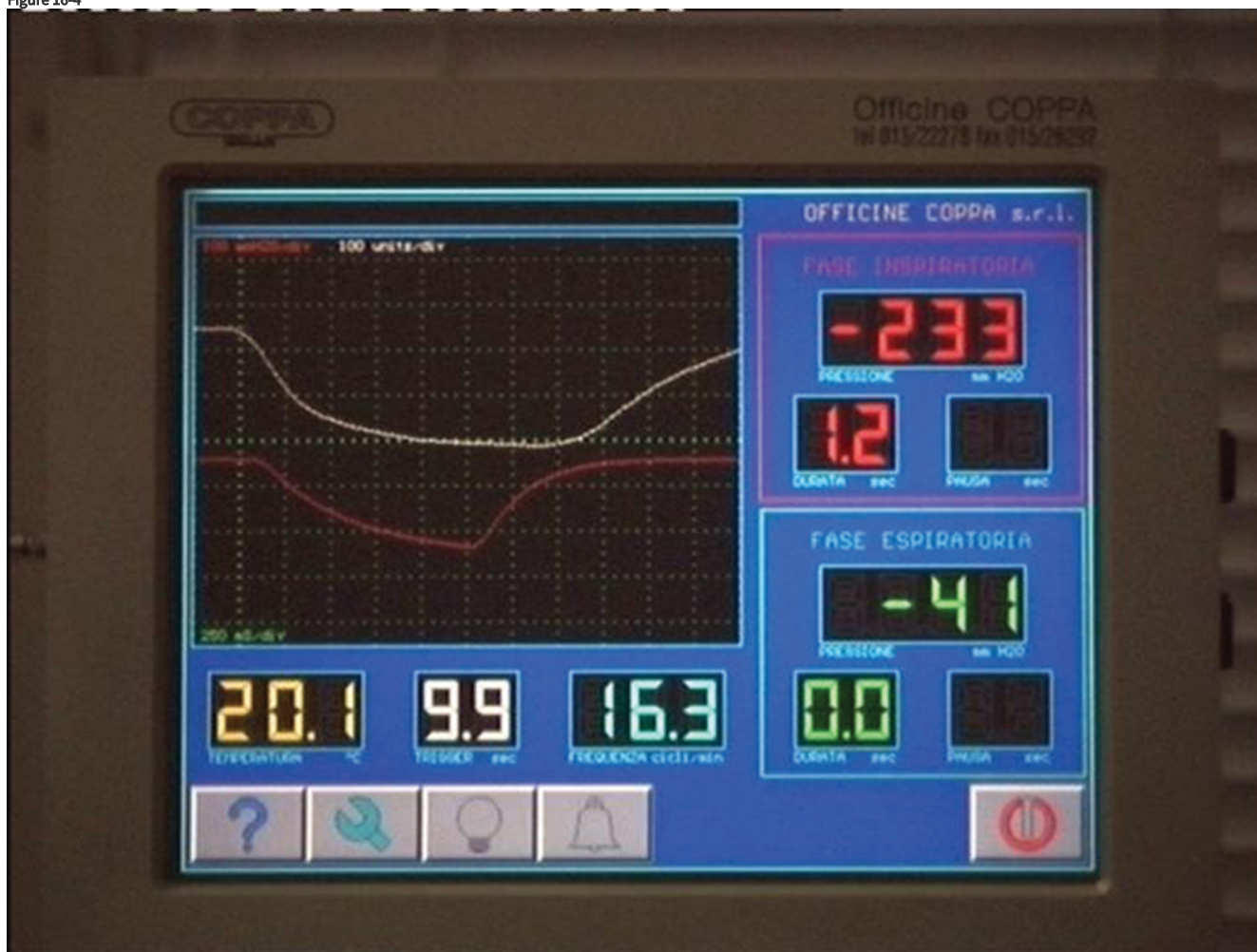
Recordings of tidal volume by Respiration Trace ($V_{T\text{ RT}}$), pleural pressure (P_{pl}), transdiaphragmatic pressure (P_{di}), and tank pressure (P_{Tank}) in a patient with an acute exacerbation of chronic obstructive pulmonary disease during assist negative-pressure ventilation provided by iron

lung.

Modes for Delivering Negative-Pressure Ventilation

Traditionally, NPV is controlled mechanical ventilation, and the device provides a fixed number of breaths per minute. If mechanical and spontaneous respiratory cycles do not match, the patient “fights” the ventilator, resulting in discomfort. Unlike PPV, during NPV, the airway opening is free; consequently, it is not possible to monitor airway pressure or flow continuously and to use these signals to trigger a mechanical breath. This patient-triggering inability may contribute to poor patient synchrony and induction of upper airway collapse secondary to the lack of coordinated activity between the upper airway and the inspiratory muscles.¹⁰ To overcome this limitation, some NPVs have incorporated patient-triggered modes using pressure change sensed via nasal prongs. There are data indicating that this technology is slow and relatively insensitive to patient inspiratory efforts.¹¹ Recently, the performance of a prototype microprocessor-based iron lung capable of thermistor triggering was evaluated.¹² The device used to trigger the iron lung was thermally sensitive and similar to that used in sleep studies, and it was activated by temperature changes caused by the onset of inspiratory airflow (Fig. 16-4). In normal subjects and patients with chronic obstructive pulmonary disease (COPD) recovering from ARF, we measured (a) the time delay between airflow onset and start of an assisted breath, (b) pressure-time product for the diaphragm per minute (PTPdi) during triggered breaths, and (c) nontriggering inspiratory efforts (Fig. 16-5). At maximum trigger sensitivity, the time delay was about 0.21 second in both groups. PTPdi was reduced markedly. Nontriggering inspiratory efforts were 1.1% and 2.3% of total breaths, respectively. Although the time delay of the thermistor trigger is longer than most recent flow-triggering and pressure-triggering systems of PPVs,¹³ the study suggests that this system permits the use of assisted NPV with an acceptable patient-ventilator interaction.

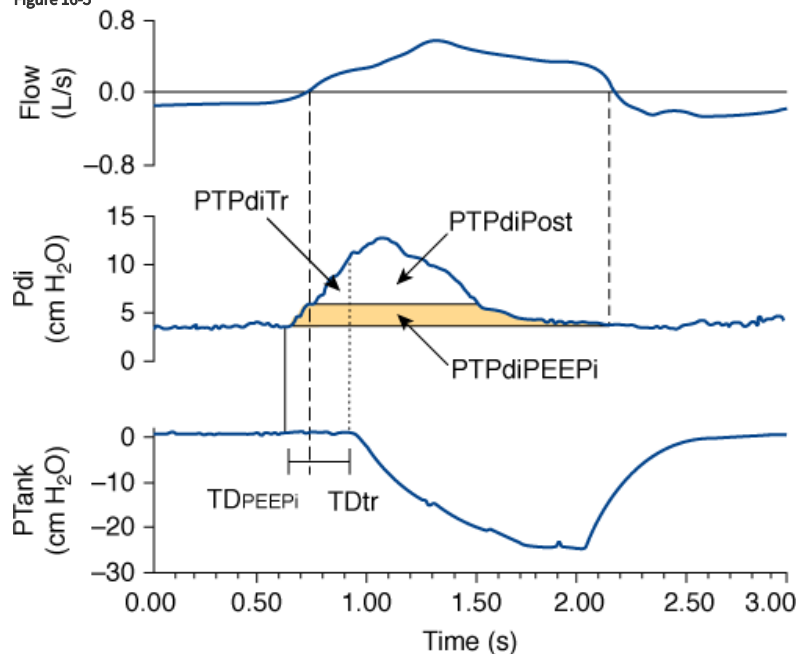
Figure 16-4



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com
Copyright © The McGraw-Hill Companies, Inc. All rights reserved.

Control panel of a microprocessor-based iron lung (Coppa CA 1001, Coppa, Biella, Italy) capable of thermistor triggering. The upper trace is the thermistor signal, and the lower trace is the tank pressure.

Figure 16-5



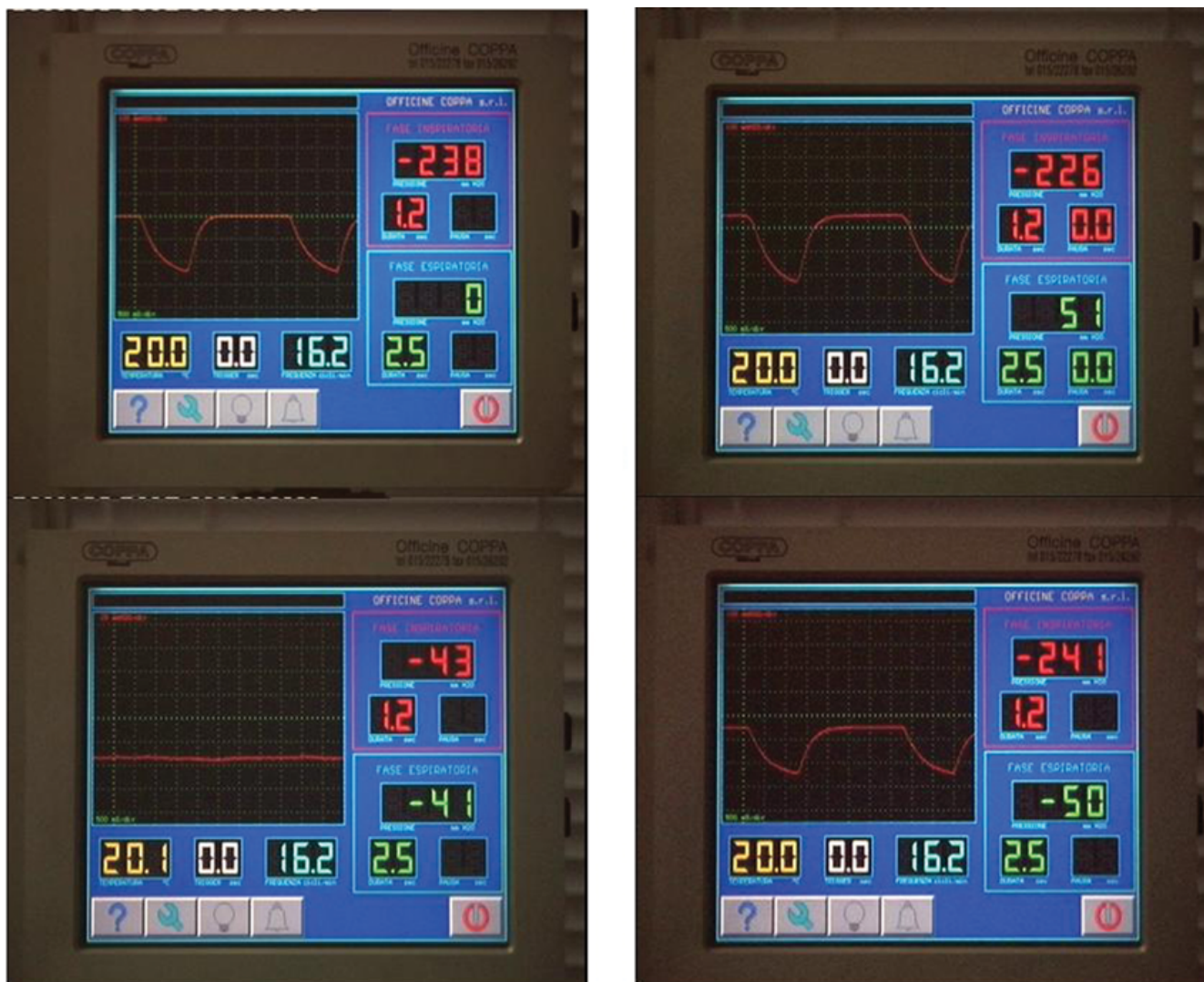
Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com

Copyright © The McGraw-Hill Companies, Inc. All rights reserved.

Recordings of flow, transdiaphragmatic pressure (P_{di}), and tank pressure (P_{tank}) in a patient with chronic obstructive pulmonary disease receiving assist NPV. The *continuous vertical line* indicates the onset of inspiratory effort, the *dashed vertical line* indicates the start of inspiratory flow, and the *dotted vertical line* indicates the start of an assisted breath. The partitioning of the pressure-time product of the diaphragm is shown: effort required to overcome intrinsic positive end-expiratory pressure ($PTP_{diPEEPi}$), effort required to trigger the assisted breath (PTP_{diTr}), and effort exerted in the post-trigger phase (PTP_{diPost}). TD_{PEEPi} and TD_{tr} indicate the time delay between the onset of inspiratory effort and the start of inspiratory flow and the time delay between the onset of inspiratory flow and the start of assisted breath, respectively. (Used, with permission, from Gorini, et al. Effect of assist negative pressure ventilation by microprocessor based iron lung on breathing effort. *Thorax*. 2002;57:258–262.)

Presently, NPV can be delivered by five modes: (a) intermittent negative pressure, (b) negative/positive pressure, (c) continuous negative pressure (CNEP), (d) negative pressure/negative end-expiratory pressure (Fig. 16-6), and (e) external high-frequency oscillation.

Figure 16-6



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com
Copyright © The McGraw-Hill Companies, Inc. All rights reserved.

Control panel of a microprocessor-based iron lung (Coppa CA 1001, Coppa, Biella, Italy) during intermittent negative pressure (*upper-left panel*), negative and/or positive pressure (*upper-right panel*), continuous negative extrathoracic pressure (*lower-left panel*), and negative pressure or negative extrathoracic end-expiratory pressure (*lower-right panel*) ventilation.

Intermittent Negative Pressure

The ventilator generates targeted subatmospheric pressure for the selected inspiratory time. Pressure around the chest wall becomes atmospheric during expiration, which occurs passively owing to the elastic recoil of the respiratory system.

Negative/Positive Pressure

The ventilator generates the preset extrathoracic subatmospheric pressure during inspiration and preset extrathoracic positive pressure during expiration. In patients with chest wall disorders, this combination has been found to increase tidal volume more than intermittent negative pressure alone by reducing the end-expiratory volume of the respiratory system.¹⁴ A useful application of this option is to assist cough and promote clearance of sputum in patients with copious secretions i.e., cystic fibrosis and bronchiectasis.⁸

Continuous Negative Extrathoracic Pressure

The ventilator provides a constant subatmospheric pressure throughout the respiratory cycle, and the patient breathes spontaneously.

Negative Pressure and Negative Extrathoracic End-Expiratory Pressure

The ventilator generates the preset subatmospheric pressure during inspiration and maintains a preset level of negative pressure throughout the expiration.

External High-Frequency Oscillation

High-frequency ventilation using a jet system can be applied via a tracheal tube or a tracheotomy. Alternatively, a cuirass can be used to apply high-frequency oscillation externally.¹⁵ The Hayek oscillator includes a cuirass, power unit, and control unit. The power unit has (a) a diaphragmatic pump, which can operate over a wide range of frequencies to generate an oscillating pressure wave, and (b) a vacuum pump, which enables the oscillation to be superimposed on a negative-pressure baseline. Peak inspiratory (up to -70 cm H₂O) and peak expiratory (up to 70 cm H₂O) pressures, frequency (8 to 999 cycles per minute), and inspiratory-to-expiratory timing (I:E) ratio (1:6 to 6:1) can be set on the control unit.

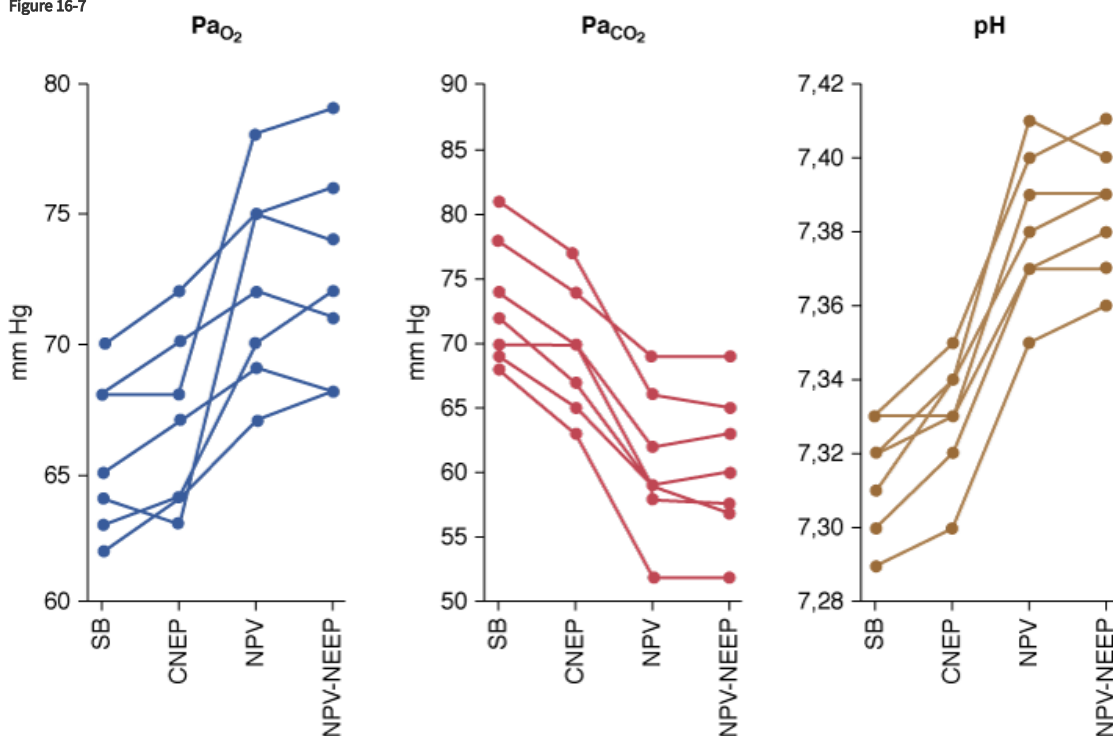
Two studies suggest that external high-frequency oscillation can provide effective ventilation in healthy subjects¹⁵ and improve end-tidal carbon dioxide (CO₂) and oxygen (O₂) saturation (SaO₂) in patients with severe stable COPD.¹⁶ In five patients with ARF, short-term application of external high-frequency oscillation improved oxygenation by 16% and reduced by 6%, compared with conventional PPV.¹⁵ In a randomized, controlled study, a 4-hour period of external high-frequency oscillation improved cardiac index and tissue perfusion in adult patients after coronary artery bypass grafting compared with conventional PPV.¹⁷

Physiologic Effects

Gas Exchange

Pioneering studies showed NPV, particularly that provided by tank ventilators, to be a highly effective form of mechanical ventilation, capable of maintaining normal arterial blood-gas tensions in patients with little or no spontaneous respiratory activity.^{18–20} Recently, it has been shown²¹ that CNEP, NPV, and NPV plus negative extrathoracic end-expiratory pressure provided by a microprocessor-based iron lung were able to improve ventilatory pattern, arterial blood gases, and pH significantly (Fig. 16-7).

Figure 16-7



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com

Copyright © The McGraw-Hill Companies, Inc. All rights reserved.

Values for, PaO₂, PaCO₂ and pH in seven patients with acute exacerbation of chronic obstructive pulmonary disease during spontaneous breathing (SB), continuous negative extrathoracic pressure (CNEP), negative pressure ventilation (NPV), and negative extrathoracic end-

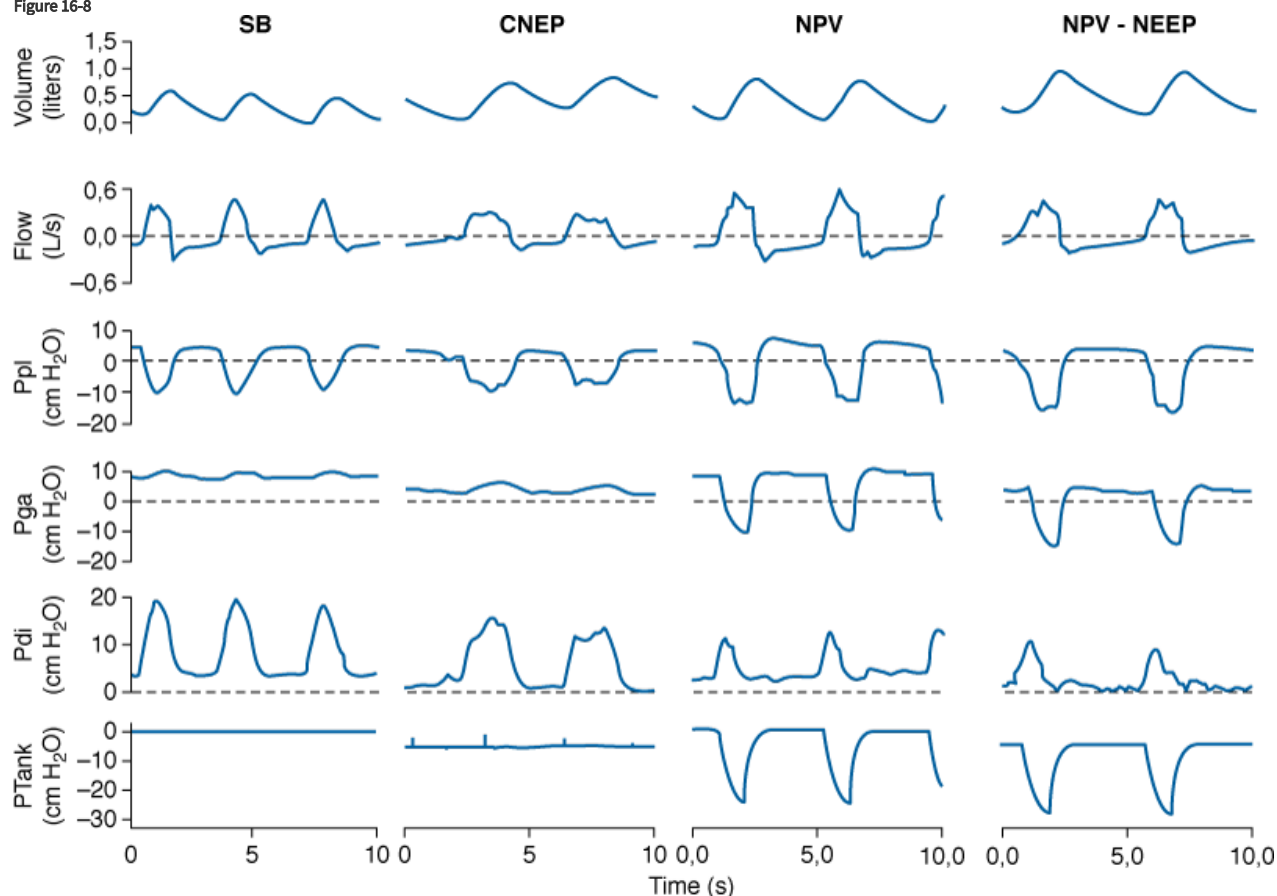
expiratory pressure added to NPV (*NPV-NEEP*). (Reprinted with permission of the American Thoracic Society. Copyright © 2012 American Thoracic Society. Used, with permission, from Gorini, et al. *Am J Respir Crit Care Med*. 2001;163:1614–1618. Official Journal of the American Thoracic Society.)

Respiratory Muscles

Two studies^{22,23} reported that in patients with acute exacerbation of COPD, NPV was effective in improving respiratory muscle strength and in decreasing PaCO_2 . In patients with chronic respiratory failure secondary to COPD and restrictive diseases, NPV reduced electrical and mechanical activity of the inspiratory muscles.^{24–28} Rodenstein et al²⁹ showed that controlled NPV provided by an old iron lung requires a short period of adaptation to obtain inspiratory muscle rest. Other studies show that when inspiratory and expiratory times are adjusted carefully to approximate the subject's spontaneous timing components, NPV results in a substantial suppression of electromyographic activity of inspiratory muscles.^{25,26} Assist-control NPV provided by a cuirass is effective in relief of dyspnea induced experimentally in normal subjects by a combination of inspiratory resistive loading and hypercapnia, probably reducing inspiratory muscle workload.³⁰

The effects of NPV provided by a microprocessor-based iron lung capable of providing CNEP and assist-control NPV on respiratory mechanics and inspiratory muscle effort in patients with COPD with ARF were evaluated by measuring the pressure-time product of the diaphragm and the electromyographic activity of parasternal muscles.²¹ Compared with spontaneous breathing, CNEP ($-5 \text{ cm H}_2\text{O}$) resulted in a significant decrease in dynamic intrinsic PEEP and pressure-time product of the diaphragm, whereas assist-control NPV caused a significant improvement in the pattern of breathing associated with a marked reduction in both pressure-time product of the diaphragm and electromyographic activity of the parasternal muscles. The application of $-5 \text{ cm H}_2\text{O}$ of negative extrathoracic end-expiratory pressure during NPV further decreased the pressure-time product of the diaphragm slightly and improved patient-ventilator interaction by reducing dynamic intrinsic PEEP and nontriggering inspiratory effort (Fig. 16-8). Reduction in diaphragmatic effort obtained during assist-control NPV²¹ is similar to that measured in patients with an acute exacerbation of COPD with pressure-support ventilation.^{31,32}

Figure 16-8



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com
Copyright © The McGraw-Hill Companies, Inc. All rights reserved.

Recordings of volume, flow, pleural pressure (P_{pl}), gastric pressure (P_{ga}), transdiaphragmatic pressure (P_{di}), and tank pressure (P_{tank}) in a patient with an acute exacerbation of COPD during spontaneous breathing (SB), continuous negative extrathoracic pressure (CNEP), negative-pressure ventilation (NPV), and negative extrathoracic end-expiratory pressure added to NPV (NPV-NEEP). The *dashed lines* indicate the level of zero in flow, P_{pl} , P_{ga} , and P_{di} recordings. (Reprinted with permission of the American Thoracic Society. Copyright © 2012 American Thoracic Society. Used, with permission, from Gorini, et al. *Am J Respir Crit Care Med*. 163:1614–1618. Official Journal of the American Thoracic Society.)

Upper Airway

The application of NPV during sleep in normal subjects,³³ and in patients with chronic respiratory failure secondary to COPD¹⁰ and neuromuscular disorders,^{34,35} may result in the development of recurrent episodes of apnea and hypopnea, as well as altered sleep quality.^{10,34} However, a recent controlled study in patients with neuromuscular disorders showed that NPV resulted in a general improvement in sleep quality and oxygen saturation.³⁶ It has been reported that in normal subjects during voluntary respiratory muscle relaxation, NPV caused a decrease in the caliber of the upper airway at the glottic or supraglottic level.³⁷ In normal, awake subjects, the glottis width did not decrease with the increase in negative pressure applied to the chest wall.³⁸ Consequently, the increasing level of NPV resulted in progressive increases in tidal volume and minute ventilation. On the contrary, control positive-pressure ventilation provided by mask caused inspiratory adduction of the vocal cords, which reduced the tidal volume effectively reaching the lungs.^{39,40}

The mechanisms of upper airway obstruction observed with NPV at supraglottic level have not been elucidated fully. During spontaneous breathing, activation of the pharyngeal and laryngeal muscles precedes activation of the inspiratory muscles, resulting in stiffening of the upper airway walls. When NPV is applied during sleep or in completely relaxed subjects, this coordinated respiratory muscle activity may be abolished. Consequently, the subatmospheric pressure developed in the upper airway during inspiration may result in their collapse.^{5,10,34,37} Upper airway obstruction has been reported in two of ten patients with acute-on-chronic respiratory failure during treatment with NPV⁴¹ and was the reason for NPV failure in 16% of patients in a large prospective cohort study.⁴²

Lower Esophageal Sphincter

NPV may induce a lower esophageal sphincter dysfunction in healthy subjects⁴³ and in patients with COPD.⁴⁴ This dysfunction may cause regurgitation of stomach contents and expose patients to the risk of aspiration. It can be prevented completely with metoclopramide.⁴⁴

Cardiovascular

The hemodynamic effects of mechanical ventilation are complex and are the result of changes in intrathoracic pressure and lung volume, which independently can affect the determinants of cardiovascular performance.^{45–47} The increase in intrathoracic pressure caused by PPV decreases both venous return to the right ventricle and left-ventricular afterload. The net effect of this reduction depends on the cardiac function. When left-ventricular function is impaired, cardiac output increases in response to the rise in intrathoracic pressure because the decrease in left-ventricular afterload secondary to the reduction in transmural pressure has a greater effect than the decrease in venous return.⁴⁸ When left-ventricular function is normal, the increase in intrathoracic pressure reduces cardiac output because the decrease in venous return has more effect than the decrease in left-ventricular afterload.⁴⁹ In clinical situations such as hypovolemia, septic shock, and gas trapping associated with airflow obstruction, the reduction in cardiac output is more relevant.⁵⁰

Although the hemodynamic effects of NPV have not been studied extensively,⁵¹ the effects are assumed to be the opposite of those of PPV, that is, more physiologic and more likely to maintain a normal cardiac output. The exposure of the entire body (except for the airway opening) to NPV by tank ventilators, however, results in the same hemodynamic effects as occurs with PPV.⁵² These effects occur because intrathoracic pressure actually is raised relative to body surface pressure, reducing the gradient for venous return. This consequence is not seen when NPV is confined to the thorax and upper abdomen using a cuirass or poncho-wrap.^{53,54} Unlike tank ventilators, these machines selectively decrease intrathoracic pressure so that right-atrial pressure becomes more negative relative to the rest of body, potentially enhancing the gradient for venous return. In an animal model of acute lung injury, NPV plus negative extrathoracic end-expiratory pressure, provided by poncho-wrap, resulted in a similar improvement in gas exchange and in higher cardiac output compared with PPV plus PEEP.⁵³

In anaesthetized dogs, PPV plus PEEP and NPV plus negative extrathoracic end-expiratory pressure applied by iron lung had a similar effect on cardiac output, whereas the latter was higher with NPV plus negative extrathoracic end-expiratory pressure applied by poncho-wrap compared with the other two modes.⁵⁴

NPV provided by cuirass does not induce adverse hemodynamic effects in stable patients with COPD,⁵⁵ whereas a significant reduction in cardiac output has been reported with mask ventilation with PEEP in patients with COPD, both when stable⁵⁶ and during acute exacerbations.⁵⁷ Short-term studies have compared the effects of CNEP provided by cuirass⁵⁸ or poncho-wrap^{59,60} with those of PEEP in intubated patients with acute lung injury receiving volume-controlled ventilation. CNEP was adjusted to obtain the same change in transpulmonary pressure⁶⁰ or functional residual capacity⁵⁸ as with PEEP. The combination of volume-controlled ventilation with CNEP, compared with volume-controlled ventilation with PEEP, resulted in significant increases in oxygen delivery and cardiac index, whereas arterial oxygen content and PaO_2/FI_2 ratio did not differ between the two modes of ventilation.^{58–60} Further studies are required to define the feasibility of long-term clinical treatment with CNEP during controlled ventilation via endotracheal intubation and its effects on clinical outcomes. Children with congenital heart diseases, submitted to right-sided heart surgery with Fontan-type procedures, have unique cardiopulmonary physiology. In the absence of a right ventricle, pulmonary blood flow, the major determinant of cardiac output, is exquisitely sensitive to changes in intrathoracic pressure. In these patients, NPV, provided by cuirass, markedly increased pulmonary blood flow and cardiac output compared with volume-controlled PPV.^{61,62}

Recently, the effects of CNEP applied to the chest through a cuirass during PPV in synchronized intermittent mandatory ventilation mode were evaluated in twenty patients with normal left-ventricular ejection fraction 2 hours after coronary artery bypass graft. With CNEP and synchronized intermittent mandatory ventilation, the stroke-volume index and cardiac index were significantly increased compared to ventilation with synchronized intermittent mandatory ventilation and PEEP. CNEP also reduced venous pressure and wedge pressure. Although the improvement in cardiac index and stroke volume was modest, the application of CNEP during PPV may have a greater benefit in patients with reduced ventricular function.⁶³

Rationale, Advantages, and Limitations

Rationale

NPV can be used to widen the field of application of noninvasive ventilator techniques. It is well known that compared with standard medical treatment, mask ventilation reduces the need for endotracheal intubation^{64–66} and reduces hospital mortality^{64–67} in selected patients with an acute exacerbation of COPD. A randomized study comparing mask with conventional mechanical ventilation in patients with exacerbations of COPD who failed medical treatment has shown that mask ventilation avoided endotracheal intubation in 48% of patients.⁶⁸ Mask ventilation, however, is not without its problems, and failure rates of 7% to 50% are reported.⁶⁹ Severe respiratory acidosis^{70–72} and illness at presentation,^{70,73} excessive airway secretions,⁷³ and inability to minimize the amount of air leakage⁷³ are major factors associated with failure of this technique.

In clinical studies, NPV has been used successfully in patients with severe respiratory acidosis or impaired level of consciousness.^{74–76} During NPV, the airway opening is free, unlike in PPV, and consequently, performing bronchial aspiration or fiberoptic bronchoscopy to remove excessive airway secretions is easy. Finally, NPV can be used in patients who cannot tolerate a mask because of facial deformity, or as a rescue therapy to avoid endotracheal intubation in those in whom mask ventilation fails. Routine implementation of noninvasive PPV in critically ill patients with acute exacerbations of COPD or severe cardiogenic pulmonary edema was associated with improved survival and reduction of nosocomial infection.⁷⁷ For these reasons, mask ventilation is recommended as the standard method of ventilator support for exacerbations of COPD; invasive mechanical ventilation is regarded as second-line rescue therapy when mask ventilation fails.⁷⁸

To verify the hypothesis that using both noninvasive mask and iron-lung ventilation should further reduce the need for endotracheal intubation in patients with acute on chronic respiratory failure, a prospective cohort study was carried out in 258 consecutive patients⁴²: 77% of the patients were treated exclusively with noninvasive ventilation (40% with NPV, 23% with mask, and 14% with the sequential use of both) and 14% with invasive ventilation. In patients in whom NPV or mask ventilation failed, sequential use of the alternative technique allowed a significant reduction in the failure of noninvasive mechanical ventilation (from 23.4% to 8.8%, $p=.002$, and from 25.3% to 5%, p

=.0001, respectively). Overall hospital mortality (21%) was lower than that estimated by Acute Physiology and Chronic Health Evaluation (APACHE) II score (28%). This study shows that use of NPV and mask ventilation made it possible to avoid endotracheal intubation in the vast majority of unselected patients with acute-on-chronic respiratory disorders needing ventilator support.⁴² Another recent study reported that using both modalities of noninvasive ventilation in patients with acute-on-chronic respiratory failure, the total rate of success in avoiding endotracheal intubation was 81.6%.⁷⁹

Advantages

As with other modalities of noninvasive ventilation, the major advantage of NPV is the avoidance of endotracheal intubation and its related complications¹⁻⁴ while preserving physiologic functions such as speech, cough, swallowing, and feeding. Moreover, because the airway opening is free, airway suction and therapeutic and diagnostic maneuvers by fiberoptic bronchoscopy are performed more easily during NPV⁴¹ than during mask ventilation.

Limitations

The following limitations should be considered when treating patients with NPV. First, the lack of upper airway protection, as with all modalities of noninvasive ventilation, may result in aspiration, especially in unconscious patients. Second, upper airway obstruction may occur or be enhanced in unconscious patients, patients with neurologic disorders that cause bulbar dysfunction, and patients with obstructive sleep apnea syndrome.^{10,33-35} This effect compromises the effectiveness of ventilation with NPVs and requires shifting to PPV. In unconscious patients with normal bulbar function, the positioning of an oropharyngeal airway can minimize the risk of airway collapse. Third, tank and wrap ventilators restrict patients to the supine position, which may induce muscular back and shoulder pain. Fourth, severe obesity and kyphoscoliosis often compromise the possibility to put the patients in an iron lung or other NPV.

The incidence of complications and side effects of NPV delivered by iron lung was evaluated recently in 153 patients with acute-on-chronic respiratory failure⁴² (Table 16-1). While NPV outcomes, such as complications and mortality rate, are within the ranges reported for noninvasive PPV,⁸⁰ it must be stressed that there are some difficulties in introducing this modality in the vast majority of ICUs, where mask ventilation is preferred as a noninvasive means of ventilation.^{81,82} The main reasons are that the iron lung is cumbersome and needs a large amount of space, and most caregivers presently have little or no experience with NPV, rather than problems associated with NPV per se.

Table 16-1: Side Effects and Complications during Negative Pressure Ventilation in 153 Patients Treated with Iron Lung for Acute-on-Chronic Respiratory Failure

	Number of Patients (%)
Patients with complications	38 (25%)
Upper airway obstruction	24 (16%)
Large air leaks	0 (0%)
Back pain	8 (5%)
Claustrophobia	17 (11.4%)
Gastric insufflation	2 (1.3%)
Major complications	
Patients with complications	3 (2%)
Pneumonia	1 (0.6%)
Pneumothorax	0 (0%)
Gastrointestinal bleeding	3 (2%)

Source: Used, with permission, from Gorini et al.⁴²

Indications and Contraindications

Acute Respiratory Failure

Chronic Obstructive Pulmonary Disease

Large studies with different experimental designs have reported the effectiveness of NPV in the treatment of patients with acute-on-chronic respiratory failure caused mainly by COPD (Table 16-2).^{42,75,79,83–89} The effectiveness of NPV provided by iron lung in patients with hypercapnic encephalopathy was evaluated retrospectively in 150 consecutive patients (79% with COPD).⁷⁴ On admission, severe hypoxemia ($\text{PaO}_2 = 56 \pm 22$ mm Hg) and hypercapnia ($\text{PaCO}_2 = 112 \pm 21$ mm Hg) with respiratory acidosis ($\text{pH } 7.13 \pm 0.13$) were present, Glasgow coma score ranged from 3 to 8, and the mean APACHE II score was 31.6 ± 5.3 . The failure rate of NPV (death or need for endotracheal intubation) was forty-five of 150 (30%); the observed mortality rate was 24% versus 67.5% predicted mortality based on APACHE II. Nine patients (6%) required intubation because of a lack of control of the airway. In recent years, the effectiveness of NPV for the treatment of acute-on-chronic respiratory failure in patients with COPD has been confirmed in two case-control studies^{75,87} and in two randomized, prospective control studies^{88,89} (Table 16-2).

Table 16-2: Clinical Studies on the Effects of NPV in Patients with Acute-on-Chronic Respiratory Failure

Authors	Experimental Design	Number of Patients	COPD (%)	Setting	Ventilator	Success* (%)
Gunella et al ⁸³	RS	560	84	Respiratory ICU	Iron lung	90
Corrado et al ⁸⁴	RS	2564	78	Respiratory ICU	Iron lung	90
Corrado et al ⁸⁵	RS	105	100	Respiratory ICU	Iron lung	89
Corrado et al ⁷⁴	RS	150	79	Respiratory ICU	Iron lung	70
Todisco et al ⁷⁹	PCS	152	72	Respiratory ICU	Iron lung	84
Gorini et al ⁴²	PCS	258	70	Respiratory ICU	Iron lung	77
Corrado et al ⁷⁵	CCS	66	100	Respiratory ICU	Iron lung	77 vs. 73
	(NPV vs invasive MV)					
Corrado et al ⁸⁷	CCS	106	100	Respiratory ICU	Iron lung	79 vs. 75
	(NPV vs. mask ventilation)					
Corrado et al ⁸⁸	RCS	44	100	Respiratory ICU	Iron lung	82 vs. 73
	(NPV vs. invasive MV)					
Corrado et al ⁸⁹	RCS	141	100	Respiratory ICU	Iron lung	87 vs. 68
	(NPV vs. mask ventilation)					

CCS, case-control study; PCS, prospective cohort study; RCS, prospective, randomized, controlled study; RS, retrospective study.

*Avoiding endotracheal intubation or death.

Neuromuscular Disorders

Few uncontrolled studies have investigated the effect of NPV in the treatment of neuromuscular patients with ARF. NPV, provided by iron lung⁹⁰⁻⁹² or pneumowrap,⁹³ was successful in avoiding endotracheal intubation and in facilitating weaning from invasive ventilation in small groups of patients. In a retrospective study, we reported the effects of NPV provided by iron lung in the treatment of fifteen neuromuscular patients with ARF.⁹⁴ On admission, all patients exhibited severe hypoxemia ($\text{PaO}_2 = 37.6 \pm 12.4$ mm Hg) and hypercapnia ($\text{PaCO}_2 = 88.2 \pm 20.4$ mm Hg) with uncompensated respiratory acidosis ($\text{pH} 7.25 \pm 0.08$). The treatment was successful in twelve of fifteen patients (80%). Although these reports suggest that NPV can be effective in the treatment of ARF in patients with neuromuscular diseases, prospective, controlled studies are needed to clarify the impact of noninvasive ventilation on clinical outcome in these patients.

Pediatric Diseases

During the 1970s and 1980s, several uncontrolled^{95–98} and controlled^{99,100} studies showed that NPV was effective in the management of the neonatal respiratory distress syndrome. More recently, Samuels et al¹⁰¹ performed a prospective, randomized, controlled trial over a period of 4 years in 244 neonates comparing CNEP (–4 to –6 cm H₂O) with standard therapy that included continuous positive airway pressure of 4 cm H₂O. They found that need for intubation was slightly less with CNEP than with standard therapy (86% vs. 91%).

Infants with ARF who fail to respond to conventional ventilation are considered elective candidates for extracorporeal membrane oxygenation. In 1989, an uncontrolled study reported that the use of CNEP administered by a tank ventilator in conjunction with invasive ventilation was successful in five neonates suffering from respiratory failure and persistent pulmonary hypertension, thus avoiding the use of extracorporeal membrane oxygenation.¹⁰² The benefit of combining invasive ventilation and CNEP in these patients has been confirmed by the same group in a crossover prospective, randomized study.¹⁰³ Patients treated with CNEP showed a greater increase in than did those treated with PEEP without any increase in morbidity.

Chronic Respiratory Failure

Chronic Obstructive Pulmonary Disease

There is no evidence that long-term NPV treatment can improve respiratory muscle function, exercise endurance, quality of life, and survival in patients with severe COPD.^{28,104–108} It is important to stress that until now, no other ventilator technique has been shown to be effective in the long-term treatment of stable patients with COPD.

Restrictive Disorders

Many studies, although uncontrolled, have shown that NPV can be used successfully for long-term home ventilation in patients with restrictive ventilatory impairment secondary to neuromuscular and chest wall diseases.^{109–114} NPV devices, however, are more cumbersome and difficult to use than recent home PPVs. In patients who need discontinuation of mask ventilation secondary to intolerable nasal irritation or upper airway congestion,¹¹⁵ and in those with other clinical condition, such as facial deformity or lack of teeth, NPV should be used as an alternative to mask ventilation.

Other Applications

Negative Pressure Ventilation during Rigid Bronchoscopy

Some studies report that in patients submitted to interventional rigid bronchoscopy under general anesthesia, NPV,^{116–118} compared with spontaneous assisted ventilation, was able to prevent apnea and significant intraoperative derangement of acid–base balance, and was associated with a reduction in administration of opioids and shortened recovery time.

Comparison with Other Modes

Negative-Pressure Ventilation versus Invasive Mechanical Ventilation

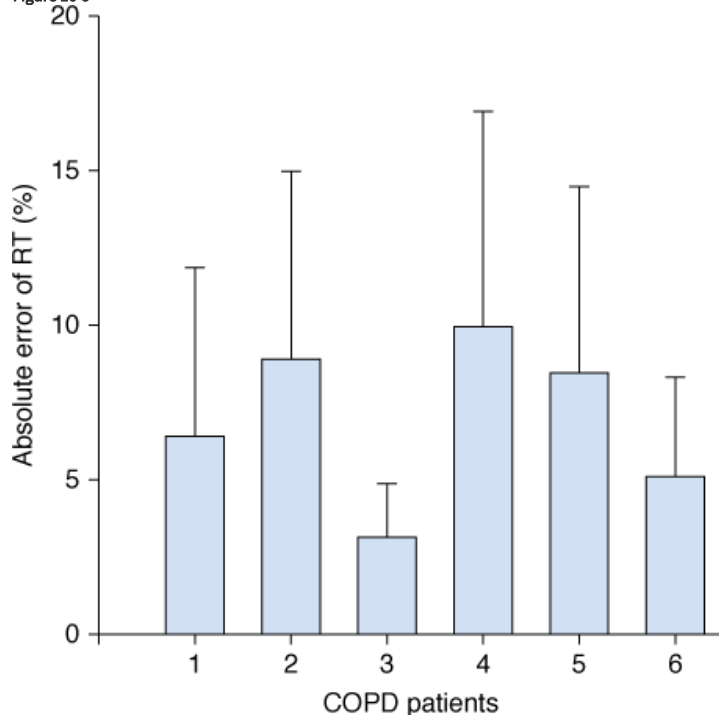
Two studies have compared the effects of NPV and conventional invasive mechanical ventilation (IMV) in patients with COPD and severe ARF.^{75,88} A retrospective case-control study, carried out in sixty-six patients,⁷⁵ showed that NPV was associated with a similar mortality rate to that of IMV (23.1% and 26.9%, respectively) and a reduction in duration of ventilation in survivors. These findings were recently confirmed in a prospective, randomized, controlled study.⁸⁸ Forty-four patients with an exacerbation of COPD and severe respiratory acidosis (mean pH 7.20 ± 0.04) were assigned either to iron-lung ventilation (twenty-two patients) or IMV (twenty-two patients). Primary end points were improvement in gas exchange and complications related to mechanical ventilation. Compared with baseline, NPV and IMV induced a similar and significant improvement in, PaO₂/FI_{O2}, PaCO₂ and pH after 1 hour and at discontinuation of treatments. Among patients treated with NPV, four (18.2%) needed endotracheal intubation. Major complications tended to be more frequent in patients treated with IMV than in those treated with NPV (27.3% vs. 4.5%), whereas mortality rate was similar (27.3% vs. 18.2%). Ventilator-free days and length of hospital stay were significantly lower in the iron-lung group than in the IMV group.

Negative-Pressure Ventilation versus Noninvasive Positive-Pressure Ventilation

A direct comparison between the two noninvasive ventilator techniques in the treatment of patients with COPD in ARF was reported in a retrospective case-control study involving fifty-three pairs of patients, well matched for age, sex, causes of ARF, APACHE II score, and pH, who were treated with iron-lung and mask ventilation.⁸⁷ Treatment failure (death and/or need for endotracheal intubation) was 20.7% in the NPV group and 24.5% in the mask ventilation group. Duration of mechanical ventilation (29.6 ± 28.6 vs. 62.3 ± 35.7 hours) and length of hospital stay (10.4 ± 4.3 vs. 15 ± 5.2 days) were significantly lower in patients treated with NPV than in those treated with mask ventilation.

This finding was recently confirmed by a multicenter prospective randomized crossover study⁸⁹ carried out in 141 patients (seventy assigned to NPV and seventy-one to mask ventilation). To establish the failure of the technique employed as first-line therapy, major and minor criteria for endotracheal intubation were used. When a major criterion was fulfilled, endotracheal intubation was promptly performed. When at least two minor criteria were met, patients were shifted from one technique to the other. When used as first-line therapy, the success of NPV (87%) was significantly greater ($P = 0.01$) than mask ventilation (68%) because of the number of patients who met minor criteria for endotracheal intubation (Fig. 16-9). After the shift between the techniques, however, the need of endotracheal intubation and hospital mortality was similar in the two groups. The total rate of success using both techniques increased from 77.3% to 87.9% ($P = 0.028$) (Fig. 16-9). These data show that when the two techniques are combined, endotracheal intubation can be avoided in a high percentage of cases.

Figure 16-9



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com

Copyright © The McGraw-Hill Companies, Inc. All rights reserved.

Error of inductance plethysmography (Respirace, RT) in estimating tidal volume during NPV in patients with an acute exacerbation of COPD. The error of the Respirace was assessed as $(V_{T, \text{Respirace}} - V_{T, \text{pneumotachograph}}) / V_{T, \text{pneumotachograph}} \times 100$. Individual values are shown.

Variations in Delivery among Ventilator Brands

Table 16-3 reports the characteristics of modern negative-pressure pumps. A study by Smith et al⁹ on five NPV pumps, using a lung model, showed that tidal volume during NPV is related both to target pressure and the pressure waveform generated by the pump. For the same target, pumps generating a square wave and produce a tidal volume up to 30% greater than pumps that generate a half sine wave.⁹ Most modern NPV pumps respond rapidly to changing leaks to maintain the preset pressure⁹ and allow independent setting of the pressure to be delivered during inspiration and expiration, as well as inspiratory and expiratory duration.

Table 16-3: Characteristics of Some Negative-Pressure Ventilator Pumps

Model	Inspiratory Pressure (cm H ₂ O)	Expiratory Pressure (cm H ₂ O)	CNEP (cm H ₂ O)	T _I (s)	T _E (s)	Ventilation Modes	Inspiratory Trigger	Pump	Alarms
Iron Lung Coppa CA 1001	0 to -80	-30 to +80	0 to -30	0.4–9.9	0.4–9.9	C, A/C, CNEP	Thermistor	Inside the model	P max, MF
Porta Lung Emerson 33-CR	0 to -90	—	Yes	Up to 5	*	C, A/C, A, CNEP	Pressure	Outside	No
Porta Lung Life Care NEV-100, Respironics	-5 to -100	-30 to +30	-5 to -30	0.5–5	†	C, A/C, C+S, A/C+S	Pressure	Outside	Pmin, MF
New Negavent DA-3 plus mod Pegaso V, Dima Italia	-5 to -99	-25 to +99	-5 to -99	‡	‡	C, A/C, CNEP	Pressure	Outside	Pmax, Pmin
Hayek RTX, Medivent International	Up to -50	Up to +50	Yes	§		C, RS, CNEP		Outside	

A, assist; A/C, assist/control; CNEP, continuous negative extrathoracic pressure; C, control; I:E, inspiratory-to-expiratory ratio; MF, machine failure; Pmax, maximum pressure; Pmin, minimum pressure; RS, respiratory synchronized; S, sigh; T_E, expiratory time; T_I, inspiratory time.

*Rate 0 to 49 cycles/min, I:E ratio: 1:3 to inverse ratio.

†Rate 4 to 60 cycles/min, I:E ratio: 1:0.5 to 1:29.

‡Rate 1 to 50 cycles/min, I:E ratio 1:9.9 to 5:1.

§Rate 6 to 1200 cycles/min, I:E ratio: 1:6 to 6:1.

Adjustments at the Bedside

Guidelines concerning the bedside management of patients ventilated with NPV are scarce.^{6,8} The effectiveness of NPV depends on strict supervision by well-trained nurses and physiotherapists with considerable expertise in this technique. Proper fit of the airtight seal around the neck and correct position of the head and neck rest are very important in optimizing comfort and preventing kinking and upper airway obstruction.⁶

Values ranging from -30 to -40 cm H₂O have been recommended for the setting of negative pressure during inspiration.⁶ In our experience, the setting of pressures and timing depends on the clinical condition and chest wall impedance of patients. In patients with severe respiratory acidosis and hypercapnic encephalopathy, we used control NPV with a frequency of 12 to 15 cycles per minute and an I:E ratio of 1:3 to 1:4, whereas assist-control NPV is used preferentially in patients with tachypnea and respiratory distress to facilitate patient-ventilator synchrony. Inspiratory negative pressure is set initially at values ranging from -30 to -40 cm H₂O and then adjusted in each patient to obtain a tidal volume of about 6 mL/kg.⁸⁷

Positive expiratory pressures of 10 to 15 cm H₂O are set to assist cough in patients with excessive secretions⁸ or to increase tidal volume in restrictive disorders;¹⁴ negative extrathoracic expiratory pressure is set to counterbalance intrinsic PEEP in patients with COPD (usually ranging from -4 to -6 cm H₂O)¹² or to improve oxygenation in patients with severe hypoxemia (ranging from -5 to -10 cm H₂O). Arterial

blood gases are checked approximately 30 minutes after mechanical ventilation starts and after any setting changes. During NPV, oxygen is supplemented by nasal cannula, venturi mask, or mask with reservoir. The flow is adjusted to keep O₂ saturation usually between 92% and 94%. Bronchodilators also can be administered by metered-dose inhaler or by aerosol.

Troubleshooting

Many problems must be considered when treating acute patients with NPVs.

Setting in Control Mode

The old iron-lung models have only this mode of ventilation, which may cause patients to “fight the ventilator,” resulting in discomfort and ineffective ventilation. In order to “capture” the patient to the superimposed frequency of the machine, we can operate mainly on the timing. Table 16-4 reports some examples of iron-lung settings in the control mode to overcome this concern.

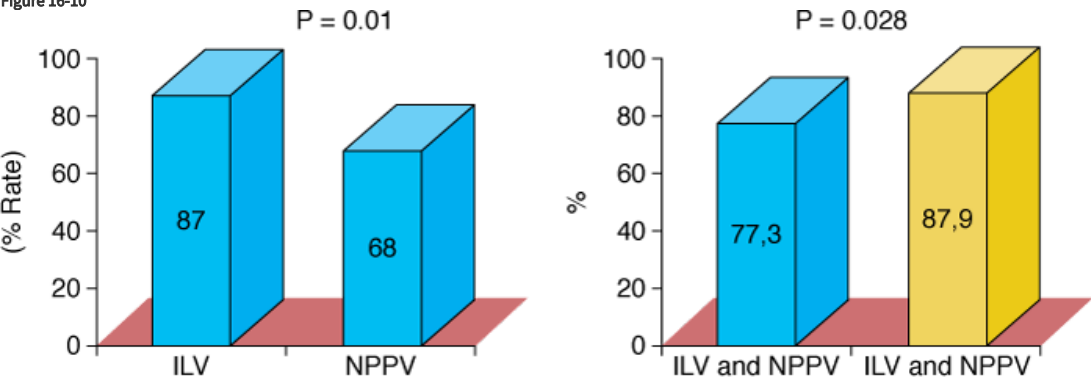
Table 16-4: Typical Setting of Iron Lung Used in Control Mode for the Treatment of Patients with Acute-on-Chronic Respiratory Failure

- Negative inspiratory pressure: Usually from -30 to -40 cm H₂O at first, and then adjusted so as to obtain a tidal volume of approximately 6 mL/kg.
- Expiratory positive pressure: From +10 to +15 cm H₂O in patients with excessive secretions to assist cough or to improve patient-ventilator synchrony.
- The timing must be adjusted some breaths below patient’s spontaneous rate.
- Typical setting for a ventilatory rate of 15 cycles/min: T_I = 1.2 s; T_E = 2.8 s; (T_I/T_{TOT} = 30%).
- Typical setting for a ventilatory rate of 27 cycles/min: T_I = 0.8 s; T_E = 1.4 s; (T_I/T_{TOT} = 36%); when patient is “captured,” respiratory rate is reduced progressively.

Monitoring of Tidal Volume

During NPV, the patient airway opening is free, which is a disadvantage for continuous monitoring of tidal volume. Currently, tidal volume is monitored intermittently using a Wright spirometer connected to a face mask. Continuous monitoring can be achieved with the use of inductance plethysmography (Respirace), which permits an indirect evaluation of tidal volume. Recently, we compared tidal volume measured by a pneumotachograph and by Respirace in patients with COPD in acute exacerbation treated by iron lung. The error of the Respirace, assessed as (V_{T,Respirace} - V_{T,pneumotachograph})/V_{T,pneumotachograph} × 100, ranged from 3% to 10% (Fig. 16-10). These observations suggest that inductance plethysmography may be a useful tool, allowing continuous estimate of tidal volume during NPV.

Figure 16-10



Source: Tobin MJ: Principles and Practice of Mechanical Ventilation, 3rd Edition: www.accessanesthesiology.com
Copyright © The McGraw-Hill Companies, Inc. All rights reserved.

The rate of success of iron-lung ventilation (ILV) and noninvasive positive pressure ventilation (NPPV) as first-line therapy in 141 patients presenting with an acute exacerbation of COPD. Left panel: The success of ILV (87%) was significantly greater than that of NPPV (68%) (P = 0.01). Right panel: Mean rate of success of ILV and NPPV used as first line of treatment and total rate of success of both techniques used sequentially.

Nursing Care

The major problems related to the assistance of patients with NPV by iron lung are transfer from the bed to inside the chamber of the tank ventilator and access to patients for nursing procedures during ventilation. Well-trained nurses, however, can manage both transfer, by using aids (e.g., a roll mattress and a mechanical elevator), and nursing procedures (e.g., insertion of a urinary catheter and venous lines, placement of electrocardiograph electrodes and SaO₂ probe for monitoring the level of oxygenation) easily using the portholes in the tank ventilator. We have measured the time spent by nurses in the second day after admission in patients with COPD in ARF treated with the iron lung.⁷⁶ The following procedures were considered: transfer of patients from the bed to inside the tank ventilator, measurement of tidal volume and minute ventilation by a Wright spirometer, tracheobronchial suction, arterial blood-gas sampling, drug administration, and other procedures. The total time spent by nurses for these procedures was 250 minutes. Even though there are no studies comparing nursing workload in patients submitted to NPV and IMV, Nava et al¹¹⁹ have reported that nursing workload in patients treated with IMV was 527.5 ± 51.1 minutes in the first 48 hours after admission.

Important Unknowns

It is completely unknown if NPV may play a role in the treatment of patients with acute lung injury. Physiologic studies^{58–60} have shown the advantageous hemodynamic effects of NPV applied to the chest wall in patients with acute lung injury, although studies on clinical outcomes are completely lacking. The absence of this information is particularly relevant because some physiologic studies suggest that NPV may result in reduced lung biotrauma compared with PPV.^{115,121,122} Physiologic studies^{52,54} show that NPV by iron lung has the same hemodynamic effects as PPV. Recently the effects of NPV, applied with a custom-made whole-body chamber, were compared to those of PPV in a model of acute lung injury in anesthetized, surfactant-depleted rabbits. NPV resulted in a superior oxygenation unrelated to lung perfusion and was associated with more effective inflation of lung volume during both inspiration and expiration.¹²³ The same group using dynamic computed tomography in six anesthetized rabbits with acute respiratory distress syndrome induced by repeated saline lavage, reported that NPV was associated with increased percentage of ventilated lung at mid-inspiration and midexpiration compared to PPV.¹²⁴

Future studies should determine whether iron-lung ventilation may be applied successfully to patients with cardiogenic pulmonary edema.

The Future

Inspiratory Trigger System

Sinderby et al¹²⁵ used diaphragmatic electrical activity, recorded by electrodes attached to a nasogastric tube, to trigger ventilatory support. This signal is directly related to phrenic nerve activity and probably to the output of the respiratory center; unlike airway pressure or flow, it is not affected by mechanical dysfunction (dynamic hyperinflation and intrinsic PEEP). By overcoming the problem associated with the current technology for triggering, this “neural trigger” has the potential to improve patient–ventilator interaction during both positive-pressure and negative-pressure mechanical ventilation.

Design of Tank Ventilator

Most iron lungs, although improved in design, remain cumbersome and need a large amount of space. A lighter and completely transparent chamber incorporated into the ICU bed would enable the posture of patients to be changed from supine to semirecumbent. This step should permit a major implementation of NPV in the ICU.

Potential Clinical Application

The clinical relevance of intra-abdominal pressure, which may cause organ dysfunction, is being recognized increasingly in the ICU settings.^{126,127} It has been suggested that NPV applied around the abdomen by a cuirass might be a potential tool to treat intra-abdominal hypertension.¹²⁸ This hypothesis has been investigated in a physiologic study of thirty patients admitted to an ICU.¹²⁹ Continuous NPV decreased intra-abdominal pressure from 8.7 ± 4.3 to 6 ± 4.2 mm Hg ($p < .001$). There was a further decrease in intra-abdominal pressure when more negative pressure was applied. Future studies with clinical outcome are necessary to assess the effectiveness of this technique for the treatment of increased abdominal pressure.

Summary and Conclusions

Evidence now exists that (a) NPV unloads respiratory muscles and improves gas exchange in patients with COPD and ARF; (b) volume-controlled ventilation with CNEP, compared with volume-controlled ventilation with PEEP, increases [oxygen](#) delivery and cardiac index in patients with acute lung injury and children submitted to right-sided heart surgery; (c) iron-lung ventilation, in expert hands, is as effective as mask ventilation and invasive ventilation in the treatment of acute-on-chronic respiratory failure; and (d) NPV can be used successfully for home mechanical ventilation in patients with chronic respiratory failure who are unable to tolerate mask ventilation. Furthermore, the use of both NPV and mask ventilation may avoid endotracheal intubation and its complications in most of these patients, widening the field of application of noninvasive ventilator techniques.

References

1. Tobin MJ. Mechanical ventilation. *N Engl J Med*. 1994;330:1056–1061. [[PubMed: 8080509](#)]

2. Mehta S, Hill NS. Noninvasive ventilation. *Am J Respir Crit Care Med*. 2001;163:540–577. [[PubMed: 11179136](#)]

3. Pingleton SK. Complications of acute respiratory failure. *Am Rev Respir Dis*. 1988;137:1463–1493. [[PubMed: 3059862](#)]

4. Fagon JY, Chastre J, Hance AJ, et al. Nosocomial pneumonia in ventilated patients: a cohort study evaluating attributable mortality and hospital stay. *Am J Med*. 1993;94:281–288. [[PubMed: 8452152](#)]

5. Levine S, Henson D. Negative pressure ventilation. In: Tobin MJ, ed. *Principles and Practice of Mechanical Ventilation*. New York, NY: McGraw-Hill; 1994:393.

6. Shneerson JM. Non-invasive and domiciliary ventilation: negative pressure techniques. *Thorax*. 1991;46:131–135. [[PubMed: 2014494](#)]

7. Corrado A, Gorini M, Villella G, et al. Negative pressure ventilation in the treatment of acute respiratory failure: an old noninvasive technique reconsidered. *Eur Respir J*. 1996;9:1531–1544. [[PubMed: 8836670](#)]

8. Hill NS. Clinical applications of body ventilators. *Chest*. 1986; 90:897–905. [[PubMed: 3536343](#)]

9. Smith IE, King MA, Shneerson JM. Choosing a negative pressure ventilation pump: are there any important differences? *Eur Respir J*. 1995;8:1792–1795. [[PubMed: 8586139](#)]

10. Levy RD, Cosio MG, Gibbons L, et al. Induction of sleep apnea with negative pressure ventilation in patients with chronic obstructive lung disease. *Thorax*. 1992;47:612–615. [[PubMed: 1412118](#)]

11. Aaron J, McCool FD, Benditt J, et al. Evaluation of trigger sensitivity of patient-triggered negative pressure ventilators. *Am J Respir Crit Care Med*. 1995;151:A237.

12. Gorini M, Villella G, Ginanni R, et al. Effects of assist negative pressure ventilation by microprocessor-based iron lung on breathing effort. *Thorax*. 2002;57:258–262. [[PubMed: 11867832](#)]

13. Aslanian P, Atrous S, Isabey D, et al. Effects of flow triggering on breathing effort during partial ventilatory support. *Am J Respir Crit Care Med*. 1998;157:135–143. [[PubMed: 9445291](#)]

14. Kinnear W, Petch M, Taylor G, Shneerson J. Assisted ventilation using cuirass respirators. *Eur Respir J*. 1988;1:198–203. [[PubMed: 3164275](#)]

15. Al-Saady NM, Fernando SSD, Petros AJ, et al. External high frequency oscillation in normal subjects and in patients with acute respiratory failure. *Anaesthesia*. 1995;50:1031–1035. [[PubMed: 8546281](#)]

16. Spitzer SA, Fink G, Mittelman M. External high-frequency ventilation in severe chronic obstructive pulmonary disease. *Chest*. 1993;104:1698–1701. [\[PubMed: 8252945\]](#)

17. Sideno B, Vaage J. Ventilation by external high-frequency oscillations improves cardiac function after coronary artery bypass grafting. *Eur J Cardiothorac Surg*. 1997;11:248–257. [\[PubMed: 9080151\]](#)

18. Lassen HCA. A preliminary report on the 1952 epidemic of poliomyelitis in Copenhagen with special reference to the treatment of acute respiratory insufficiency. *Lancet*. 1953;1:37–41. [\[PubMed: 13011944\]](#)

19. Lovejoy FW, Yu PNG, Nye RE, et al. Pulmonary hypertension: III. Physiological studies in three cases of carbon dioxide narcosis treated by artificial respiration. *Am J Med*. 1954;16:4–11. [\[PubMed: 13114231\]](#)

20. Marks A, Bocles J, Morganti L. A new ventilatory assister for patients with respiratory acidosis. *N Engl J Med*. 1963;268:61–7.

21. Gorini M, Corrado A, Villella G, et al. Physiologic effects of negative pressure ventilation in acute exacerbation of COPD. *Am J Respir Crit Care Med*. 2001;163:1614–1618. [\[PubMed: 11401883\]](#)

22. Corrado A, Bruscoli G, De Paola E, et al. Respiratory muscle insufficiency in acute respiratory failure of subjects with severe COPD: treatment with intermittent negative pressure ventilation. *Eur Respir J*. 1990;3:644–648. [\[PubMed: 2379575\]](#)

23. Montserrat JM, Martos JA, Alarcon A, et al. Effect of negative pressure ventilation on arterial blood gas pressure and inspiratory muscle strength during an exacerbation of chronic obstructive lung disease. *Thorax*. 1991;46:6–8. [\[PubMed: 1908138\]](#)

24. Nava S, Ambrosino N, Zocchi L, et al. Diaphragmatic rest during negative pressure ventilation by pneumowrap: assessment in normal and COPD patients. *Chest*. 1990;98:850–856.

25. Gigliotti F, Duranti R, Fabiani A, et al. Suppression of ventilatory muscle activity in healthy subjects and COPD patients with negative pressure ventilation. *Chest*. 1991;99:1186–1192. [\[PubMed: 2019176\]](#)

26. Gigliotti F, Spinelli A, Duranti R, et al. Four-week negative pressure ventilation improves respiratory function in severe hypercapnic COPD patients. *Chest*. 1994;105:87–89. [\[PubMed: 8275793\]](#)

27. Belman MJ, Soo Hoo GW, Kuei JH, et al. Efficacy of positive vs negative pressure evaluation in unloading respiratory muscles. *Chest*. 1990;98:850–856. [\[PubMed: 2119950\]](#)

28. Rochester DF, Braun NMT, Laine S. Diaphragmatic energy expenditure in chronic respiratory failure: the effect of assisted ventilation with body respirators. *Am J Med*. 1977;63:223–232. [\[PubMed: 268143\]](#)

29. Rodenstein DO, Stanescu DC, Cuttita G, et al. Ventilatory and diaphragmatic EMG responses to negative pressure ventilation in airflow obstruction. *J Appl Physiol*. 1988;65:1621–1626. [\[PubMed: 3182526\]](#)

30. Nishino T, Isono S, Ide T. Effects of negative pressure assisted ventilation on dyspnoeic sensation and breathing pattern. *Eur Respir J*. 1998;12:1278–1283. [\[PubMed: 9877477\]](#)

31. Appendini L, Patessio A, Zanaboni S, et al. Physiologic effects of positive end-expiratory pressure and mask pressure support during exacerbations of chronic obstructive pulmonary disease. *Am J Respir Crit Care Med*. 1994;149:1069–1076. [\[PubMed: 8173743\]](#)

32. Leung P, Jubran A, Tobin MJ. Comparison of assisted ventilator modes on triggering, patient effort, and dyspnea. *Am J Respir Crit Care Med*. 1997;155:1940–1948. [\[PubMed: 9196100\]](#)

33. Levy RD, Bradley TD, Newman SL, et al. Negative pressure ventilation: effects on ventilation during sleep in normal subjects. *Chest*. 1989;95:95–99. [\[PubMed: 2909361\]](#)

34. Hill NS, Redline S, Carskadon MA, et al. Sleep-disordered breathing in patients with Duchenne muscular dystrophy using negative pressure ventilators. *Chest*. 1992;102:1656–1662. [[PubMed: 1446467](#)]
-
35. Bach JR, Penek J. Obstructive sleep apnea complicating negative pressure ventilatory support in patients with chronic paralytic/restrictive ventilatory dysfunction. *Chest*. 1991;99:1386–1393. [[PubMed: 2036820](#)]
-
36. Schiavina M, Fabbri A. Intermittent negative pressure ventilation in patients with respiratory failure. *Monaldi Arch Chest Dis*. 1993;48:169–175. [[PubMed: 8518781](#)]
-
37. Sanna A, Veriter C, Stanescu D. Upper airway obstruction induced by negative-pressure ventilation in awake healthy subjects. *J Appl Physiol*. 1993;75:546–552. [[PubMed: 8226452](#)]
-
38. Glerant JC, Jounieaux V, Parreira VF, et al. Effects of intermittent negative pressure ventilation on effective ventilation in normal awake subjects. *Chest*. 2002;122:99–107. [[PubMed: 12114344](#)]
-
39. Jounieaux V, Aubert G, Dury M, et al. Effects of nasal positive-pressure hyperventilation on the glottis in normal awake subjects. *J Appl Physiol*. 1995;79:176–185. [[PubMed: 7559217](#)]
-
40. Parreira VF, Jounieaux V, Aubert G, et al. Nasal two-level positive-pressure ventilation in normal subjects. *Am J Respir Crit Care Med*. 1996;153:1616–1623. [[PubMed: 8630611](#)]
-
41. Todisco T, Eslami A, Scarcella L, et al. Flexible bronchoscopy during iron lung mechanical ventilation in nonintubated patients. *J Bronchol*. 1995;2:200–205.
-
42. Gorini M, Ginanni R, Villella G, et al. Non-invasive negative and positive pressure ventilation in the treatment of acute on chronic respiratory failure. *Intensive Care Med*. 2004;30:875–881. [[PubMed: 14735237](#)]
-
43. Marino WD, Jain NK, Pitchumoni CS. Induction of lower esophageal sphincter (LES) dysfunction during use of the negative pressure body ventilator. *Am J Gastroenterol*. 1988;83:1376–1380. [[PubMed: 3195543](#)]
-
44. Marino WD, Pitchumoni CS. Reversal of negative pressure ventilation induced lower esophageal sphincter dysfunction with metoclopramide. *Am J Gastroenterol*. 1992;87:190–194. [[PubMed: 1734696](#)]
-
45. Pinsky MR. The effects of mechanical ventilation on the cardiovascular system. *Crit Care Med*. 1990;6:663–678. [[PubMed: 2199000](#)]
-
46. Pinsky MR. The hemodynamic consequences of mechanical ventilation an evolving story. *Intensive Care Med*. 1997;23:493–503. [[PubMed: 9201520](#)]
-
47. Takata M, Robotham JL. Effects of inspiratory diaphragmatic descent on inferior vena caval venous return. *J Appl Physiol*. 1992;72:597–607. [[PubMed: 1559938](#)]
-
48. Pinsky MR, Summer WR. Cardiac augmentation by phasic high intrathoracic support in man. *Chest*. 1983;84:370–375. [[PubMed: 6617271](#)]
-
49. Cassidy SA, Wead WB, Seibert GB, et al. Geometric left ventricular responses to interaction between the lung and left ventricle: positive pressure breathing. *Ann Biomed Eng*. 1987;15:373–383.
-
50. Jardin F, Farcot JC, Boisante L, et al. Influence of positive end-expiratory pressure on left ventricular performance. *N Engl J Med*. 1981;304:387–392. [[PubMed: 7005679](#)]
-
51. Krumpe PE, Zidulka A, Urbanetti J, et al. Comparison of the effects of continuous negative external chest pressure and positive end-expiratory pressure on cardiac index in dogs. *Am Rev Respir Dis*. 1977;115:39–45. [[PubMed: 319716](#)]
-

52. Maloney JV, Whittenberger JL. Clinical implication of pressures used in the body respirator. *Am J Med Sci.* 1951;221:425–430. [PubMed: 14819018]
-
53. Skabursis M, Helal R, Zidulka A. Hemodynamic effects of external continuous negative pressure ventilation compared with those of continuous positive pressure ventilation in dogs with acute lung injury. *Am Rev Respir Dis.* 1987;136:886–891.
-
54. Lockhat D, Langleben D, Zidulka A. Hemodynamic differences between continuous positive and two types of negative pressure ventilation. *Am Rev Respir Dis.* 1992;146:677–680. [PubMed: 1519847]
-
55. Ambrosino N, Cobelli F, Torbicki A, et al. Hemodynamic effects of negative-pressure ventilation in patients with COPD. *Chest.* 1990;97:850–856. [PubMed: 2182297]
-
56. Ambrosino N, Nava S, Torbicki A, et al. Haemodynamic effects of pressure support and PEEP ventilation by nasal route in patients with stable chronic obstructive pulmonary disease. *Thorax.* 1993;48:523–528. [PubMed: 8322240]
-
57. Diaz O, Iglesia R, Ferrer M, et al. Effects of noninvasive ventilation on pulmonary gas exchange and hemodynamics during acute hypercapnic exacerbations of chronic obstructive pulmonary disease. *Am J Respir Crit Care Med.* 1997;156:1840–1845. [PubMed: 9412564]
-
58. Scholz SE, Knothe C, Thiel A, Hempelmann G. Improved oxygen delivery by positive pressure ventilation with continuous negative external chest pressure. *Lancet.* 1997;349:1295–1296. [PubMed: 9142069]
-
59. Torelli L, Zoccali G, Casarin M, et al. Comparative evaluation of the haemodynamic effects of continuous negative external pressure (CNEP) and positive end-expiratory pressure (PEEP) in mechanically ventilated trauma patients. *Intensive Care Med.* 1995;21:67–70. [PubMed: 7560477]
-
60. Borelli M, Benini A, Denkevitz T, et al. Effects of continuous negative extrathoracic pressure versus positive end-expiratory pressure in acute lung injury patients. *Crit Care Med.* 1998;26:1025–1031. [PubMed: 9635650]
-
61. Shekerdemian LS, Shore DF, Lincoln C, et al. Negative pressure ventilation improves cardiac output after right heart surgery. *Circulation.* 1996;94:49–55.
-
62. Shekerdemian LS, Bush A, Shore DF, et al. Cardiopulmonary interaction after Fontan operations: augmentation of cardiac output using negative pressure ventilation. *Circulation.* 1997;96:3934–3942. [PubMed: 9403618]
-
63. Chaturvedi RK, Zidulka AA, Goldberg P, et al. Use of negative extrathoracic pressure to improve hemodynamics after cardiac surgery. *Ann Thorac Surg.* 2008;85:1355–1360. [PubMed: 18355527]
-
64. Brochard L, Mancebo J, Wysochi M, et al. Noninvasive ventilation for acute exacerbations of chronic obstructive pulmonary disease. *N Engl J Med.* 1995;333:817–822. [PubMed: 7651472]
-
65. Kramer N, Meyer TJ, Meharg J, et al. Randomized, prospective trial of noninvasive positive pressure ventilation in acute respiratory failure. *Am J Respir Crit Care Med.* 1995;151:1799–1806. [PubMed: 7767523]
-
66. Plant PK, Owen JL, Elliott MW. Early use of non-invasive ventilation for acute exacerbations of chronic obstructive pulmonary disease on general respiratory wards: a multicentre randomised, controlled trial. *Lancet.* 2000;355:1931–1935. [PubMed: 10859037]
-
67. Bott J, Carroll MP, Conway JH, et al. Randomised, controlled trial of nasal ventilation in acute ventilatory failure due to chronic obstructive airways disease. *Lancet.* 1993;341:1555–1557. [PubMed: 8099639]
-
68. Conti G, Antonelli M, Navalesi P, et al. Noninvasive vs conventional mechanical ventilation in patients with chronic obstructive pulmonary disease after failure of medical treatment in the ward: a randomized trial. *Intensive Care Med.* 2002;28:1701–1707. [PubMed: 12447511]
-

69. Lightowler JVJ, Elliott MW. Predicting the outcome of NIV for acute exacerbations of COPD. *Thorax*. 2000;55:815–816. [PubMed: 10992530]
-
70. Ambrosino N, Foglio K, Rubini F, et al. Non-invasive mechanical ventilation in acute respiratory failure due to chronic obstructive pulmonary disease: correlates for success. *Thorax*. 1995;50:755–757. [PubMed: 7570410]
-
71. Plant PK, Owen JL, Elliott MW. Non-invasive ventilation in acute exacerbations of chronic obstructive pulmonary disease: long-term survival and predictors of in-hospital outcome. *Thorax*. 2001;56:708–712. [PubMed: 11514692]
-
72. Carlucci A, Richard JC, Wysocki M, et al. Noninvasive versus conventional mechanical ventilation: an epidemiologic survey. *Am J Respir Crit Care Med*. 2001;163:874–880. [PubMed: 11282759]
-
73. Soo Hoo GW, Santiago S, Williams A. Nasal mechanical ventilation for hypercapnic respiratory failure in chronic obstructive pulmonary disease: determinants of success and failure. *Crit Care Med*. 1994;22:1253–1261.
-
74. Corrado A, De Paola E, Gorini M, et al. Intermittent negative pressure ventilation in the treatment of hypoxic hypercapnic coma in chronic respiratory insufficiency. *Thorax*. 1996;5:1077–1082.
-
75. Corrado A, Gorini M, Ginanni R, et al. Negative pressure ventilation versus conventional mechanical ventilation in the treatment of acute respiratory failure in COPD patients. *Eur Respir J*. 1998;12:519–525. [PubMed: 9762773]
-
76. Corrado A, Gorini M. Negative pressure ventilation: is there still a role? *Eur Respir J*. 2002;20:187–197. [PubMed: 12166569]
-
77. Girou E, Brun Buisson C, Taille S, et al. Secular trends in nosocomial infections and mortality associated with noninvasive ventilation in patients with exacerbation of COPD and pulmonary edema. *JAMA*. 2003;290:2985–2991. [PubMed: 14665660]
-
78. Elliott MW. Non-invasive ventilation in acute exacerbations of chronic obstructive pulmonary disease: a new gold standard? *Intensive Care Med*. 2002;28:1691–1693. [PubMed: 12580151]
-
79. Todisco T, Baglioni S, Eslami A, et al. Treatment of acute exacerbations of chronic respiratory failure. *Chest*. 2004;125:2217–2223. [PubMed: 15189944]
-
80. Mehta S, Hill NS. Noninvasive ventilation. *Am J Respir Crit Care Med*. 2001;163:540–577. [PubMed: 11179136]
-
81. Nava S, Confalonieri M, Rampulla C. Intermediate respiratory intensive care units in Europe: a European perspective. *Thorax*. 1998;53:798–802. [PubMed: 10319065]
-
82. European Respiratory Task Force. Respiratory intermediate care units: a European survey. *Eur Respir J*. 2002;20:1343–1350.
-
83. Gunella G. Traitement de l'insuffisance respiratoire aiguë des pulmonaires chroniques avec le poumon d'acier: résultat dans une série de 560 cas. *Ann Med Phys*. 1980;2:317–327.
-
84. Corrado A, Gorini M, De Paola E, et al. Iron lung treatment of acute on chronic respiratory failure: 16 years of experience. *Monaldi Arch Chest Dis*. 1994;49:552–555. [PubMed: 7711715]
-
85. Corrado A, Bruscoli G, Messori A, et al. Iron lung treatment of subjects with COPD in acute respiratory failure: evaluation of short and long-term prognosis. *Chest*. 1992;101:692–696. [PubMed: 1541134]
-
86. Corrado A, De Paola E, Messori A, et al. The effect of intermittent negative pressure ventilation and long-term oxygen therapy for patients with COPD: a 4-year study. *Chest*. 1994;105:95–99. [PubMed: 8275794]
-
87. Corrado A, Confalonieri M, Marchese S, et al. Iron lung versus mask ventilation in the treatment of acute on chronic respiratory failure in COPD patients: a multicenter study. *Chest*. 2002;121:189–195. [PubMed: 11796450]
-

88. Corrado A, Ginanni R, Vilella G, et al. Iron lung versus conventional mechanical ventilation in acute exacerbation of COPD. *Eur Respir J*. 2004;23:419–424. [PubMed: 15065832]
-
89. Corrado A, Gorini M, Melej R, et al. Iron lung versus mask ventilation in acute exacerbation of COPD: a randomized crossover study. *Intensive Care Med*. 2009;35:648–655. [PubMed: 19020859]
-
90. Del Bufalo C, Fasano L, Quarta CC, et al. Use of extrathoracic negative pressure ventilation in weaning COPD and kyphoscoliotic patients from mechanical ventilation. *Respir Care*. 1994;39:21–29.
-
91. Garay SM, Turino GM, Goldring RM. Sustained reversal of chronic hypercapnia in patients with alveolar hypoventilation syndromes: long-term maintenance with noninvasive nocturnal mechanical ventilation. *Am J Med*. 1981;70:269–274. [PubMed: 6781339]
-
92. Libby BM, Briscoe WA, Boyce B, et al. Acute respiratory failure in scoliosis or kyphosis. *Am J Med*. 1982;73:532–538. [PubMed: 6812420]
-
93. Braun SR, Sufit RL, Giovannoni R, et al. Intermittent negative pressure ventilation in the treatment of respiratory failure in progressive neuromuscular disease. *Neurology*. 1987;37:1874–1875. [PubMed: 3683880]
-
94. Corrado A, Gorini M, De Paola E. Alternative techniques for managing acute neuromuscular respiratory failure. *Semin Neurol*. 1995; 15:84–89. [PubMed: 7638463]
-
95. Brancalari E, Gerhardt T, Monkus E. Simple device for producing continuous negative pressure in infants with IRDS. *Pediatrics*. 1973;52:128–130.
-
96. Alexander G, Gerhardt T, Brancalari E. Hyaline membrane disease: comparison of continuous negative pressure and nasal positive airway pressure in its treatment. *Am J Dis Child*. 1979;133:1156–1159. [PubMed: 389037]
-
97. Outerbridge EW, Roloff DW, Stern L. Continuous negative pressure in the management of severe respiratory distress syndrome. *J Pediatr*. 1972;81:384–391. [PubMed: 4557605]
-
98. Samuels MP, Southall DP. Negative extrathoracic pressure in the treatment of respiratory failure in infants and young children. *Br Med J*. 1989;299:1253–1257. [PubMed: 2513900]
-
99. Fanaroff AA, Cha CC, Sosa R, et al. Controlled trial of continuous negative external pressure in the treatment of severe respiratory distress syndrome. *J Pediatr*. 1973;82:921–928. [PubMed: 4573990]
-
100. Silverman WA, Sinclair JC, Gandy GM, et al. A controlled trial of management of respiratory distress syndrome in a body-enclosing respirator: I. Evaluation of safety. *Pediatrics*. 1967;39:740–748. [PubMed: 5229705]
-
101. Samuels MP, Raine J, Wright T. Continuous negative extrathoracic pressure in neonatal respiratory failure. *Pediatrics*. 1996;98:1154–1160. [PubMed: 8951269]
-
102. Sillis JH, Cvetnic WG, Pietz J. Continuous negative pressure in the treatment of infants with pulmonary hypertension and respiratory failure. *J Perinatol*. 1989;9:43–48.
-
103. Cvetnic WG, Shouptaug M, Sillis JH. Intermittent mandatory ventilation with continuous negative pressure compared with positive end-expiratory pressure for neonatal hypoxemia. *J Perinatol*. 1992;12:316–324. [PubMed: 1479457]
-
104. Zibrack JD, Hill NS, Federman EC, et al. Evaluation of intermittent long-term negative pressure ventilation in patients with severe chronic obstructive pulmonary disease. *Am Rev Respir Dis*. 1988;138:1515–1518.
-
105. Cropp AJ, Dimarco AF. Effect of intermittent negative pressure ventilation on respiratory muscle function in patients with severe COPD. *Am Rev Respir Dis*. 1987;135:1056–1061. [PubMed: 3107444]
-

106. Gutierrez M, Beroiza T, Contreras G. Weekly cuirass ventilation improves blood gases and inspiratory muscle strength in patients with chronic airflow limitation and hypercarbia. *Am Rev Respir Dis.* 1988;138:617–623. [[PubMed: 3144206](#)]

107. Celli B, Lee H, Criner G, et al. Controlled trial of external negative pressure ventilation in patients with severe chronic airflow obstruction. *Am Rev Respir Dis.* 1989;140:1251–1256. [[PubMed: 2683904](#)]

108. Shapiro SH, Ernst P, Gray-Donald K, et al. Effect of negative pressure ventilation in severe chronic obstructive pulmonary disease. *Lancet.* 1992;340:1425–1429. [[PubMed: 1360558](#)]

109. Garay SM, Turino GM, RM. Sustained reversal of chronic hypercapnia in patients with alveolar hypoventilation syndromes: long-term maintenance with noninvasive nocturnal mechanical ventilation. *Am J Med.* 1981;70:269–274. [[PubMed: 6781339](#)]

110. Curran FJ, Colbert AP. Ventilatory management in Duchenne muscular dystrophy: twelve years' experience. *Arch Phys Med Rehabil.* 1989;70:180–185. [[PubMed: 2647055](#)]

111. Jackson M, Kinnear W, King M, et al. The effects of five years of nocturnal cuirass-assisted ventilation in chest wall disease. *Eur Respir J.* 1993;6:630–635. [[PubMed: 8519371](#)]

112. Splaingard ML, Frates RC, Jefferson LS, et al. Home negative pressure ventilation: report of 20 years' experience in patients with neuromuscular disease. *Arch Phys Med Rehabil.* 1985;66:239–242. [[PubMed: 3885906](#)]

113. Schiavina M, Fabiani A. Intermittent negative pressure ventilation in patients with restrictive respiratory failure. *Monaldi Arch Chest Dis.* 1993;48:169–175. [[PubMed: 8518781](#)]

114. Baydur A, Layne E, Aral H, et al. Long term non-invasive ventilation in the community for patients with musculoskeletal disorders: 46-year experience and review. *Thorax.* 2000;55:4–11. [[PubMed: 10607795](#)]

115. Culver BH, Butler J. Mechanical influence on the pulmonary microcirculation. *Annu Rev Physiol.* 1980;42:187–198. [[PubMed: 6996579](#)]

116. Vitacca M, Natalini G, Cavaliere S, et al. Breathing pattern and arterial blood gases during Nd-YAG laser photoresection of endobronchial lesions under general anaesthesia: use of negative pressure ventilation—a preliminary study. *Chest.* 1997;112:1466–1473. [[PubMed: 9404740](#)]

117. Natalini G, Cavaliere S, Vitacca M, et al. Negative pressure ventilation vs spontaneous assisted ventilation during rigid bronchoscopy. *Acta Anaesthesiol Scand.* 1998;42:1063–1069. [[PubMed: 9809089](#)]

118. Natalini G, Cavaliere S, Seramondi V, et al. Negative pressure ventilation vs external high-frequency oscillation during rigid bronchoscopy: a controlled, randomized trial. *Chest.* 2000;118:18–23. [[PubMed: 10893353](#)]

119. Nava S, Evangelisti I, Rampulla C, et al. Human and financial costs of noninvasive mechanical ventilation in patients affected by COPD and acute respiratory failure. *Chest.* 1997;111:1631–1638. [[PubMed: 9187186](#)]

120. Uhlig S. Ventilation-induced lung injury and mechanotransduction: stretching it too far? *Am J Physiol Lung Cell Mol Physiol.* 2002;282:L893–L896.

121. Koyama S, Hildebrandt J. Air interface and elastic recoil affect vascular resistance in three zone of rabbit lung. *J Appl Physiol.* 1991; 70:2422–2431. [[PubMed: 1885435](#)]

122. Von Bethmann AN, Brasch F, Nusing R, et al. Hyperventilation reduces release of cytokines from perfused mouse lung. *Am J Respir Crit Care Med.* 1998;157:263–272.

123. Grasso F, Engelberts D, Helm E, et al. Negative pressure ventilation. Better oxygenation and less lung injury. *Am J Respir Crit Care Med.* 2008;177:412–418. [[PubMed: 18079496](#)]

124. Helm E, Talakoub O, Grasso F, et al. Use of dynamic CT in acute respiratory distress syndrome (ARDS) with comparison of positive and negative pressure ventilation. *Eur Radiol.* 2009;19:50–57. [[PubMed: 18651149](#)]

125. Sinderby C, Navalesi P, Beck J, et al. Neural control of mechanical ventilation in respiratory failure. *Nat Med.* 1999;5:1433–1436. [[PubMed: 10581089](#)]

126. Malbrain ML. Abdominal pressure in the critically ill: measurement and clinical relevance. *Intensive Care Med.* 1999;25:1453–1458. [[PubMed: 10702030](#)]

127. Bailey J, Shapiro MJ. Abdominal compartment syndrome. *Crit Care.* 2000;4:23–29. [[PubMed: 11094493](#)]

128. Bloomfield G, Saggi B, Blocher C, et al. Physiologic effects of externally applied continuous negative abdominal pressure for intra-abdominal hypertension. *J Trauma.* 1999;46:1009–1014. [[PubMed: 10372616](#)]

129. Valenza F, Bottino N, Canavesi K, et al. Intra-abdominal pressure may be decreased non-invasively by continuous negative extra-abdominal pressure (NEXAP). *Intensive Care Med.* 2003;29:2063–2067. [[PubMed: 14530858](#)]

[McGraw Hill](#)

Copyright © McGraw-Hill Global Education Holdings, LLC.

All rights reserved.

Your IP address is **132.174.254.28**

Access Provided by: Walter Reed National Military Medical Center

[Silverchair](#)